R18

Code No: 153AG

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech II Year I Semester Examinations, August/September - 2022

COMPUTER ORGANIZATION AND ARCHITECTURE

(Common to CSE, CSBS, CSIT, CSE(SE), CSE(CS), CSE(AIML), CSE(DS), CSE(N))

Time: 3 Hours **Max.Marks:75**

Answer any five questions

All questions carry equal marks

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1.a)	What is the difference between Computer Organization and Computer Architecture?	
b)	Write a short note on instruction code format.	[8+7]
2.a)	Explain the format of Register reference instructions and their functionalities.	
b)	Draw and explain the flowchart for interrupt cycle.	[8+7]
3.a)	Define the following terms: i) Control memory ii) Address sequencing	
b)	Explain about the microinstruction format with neat sketch.	[8+7]
4.a)	Explain about various addressing modes.	
b)	Briefly explain about General purpose registers and Flag registers.	[7+8]
5.a)	Explain the flow chart for addition operation with sign-magnitude data.	
b)	Perform (-25) + (-10) in binary with negative numbers in 2's complement.	[8+7]
6.a)	Explain the Booth's algorithm for signed multiplication.	
b)	Draw the flowchart for floating point division.	[8+7]
7.a)	Explain the block diagram of I/O interface.	
b)	Write a short note on Cache memory.	[8+7]
8.a)	Explain about instruction pipelining with an example.	
b)	Discuss about the serial arbitration technique.	[8+7]

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